

Investigate the transfer and transformation of energy in electrical circuits, including the role of circuit components, insulators and conductors

Learning intention

- investigate the transfer and transformation of energy in electrical circuits, including the role of circuit components, insulators and conductors.

Teach and model

- investigating different electrical conductors and insulators and examining why they may be used.
- constructing representations of electrical circuits and their components using accepted conventions.
- examining the purpose of different components such as switches and bulbs and exploring use of ammeters to measure current.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.